

CANDIDATE ELIMINATION

CODE

```
import pandas as pd

data = pd.read_csv("findsalgorithm.csv")

print("Training Data:\n")

print(data)

concepts = np.array(data.iloc[:, :-1])

target = np.array(data.iloc[:, -1])

n = concepts.shape[1]

S = concepts[0].copy()

G = [["?" for i in range(n)] for j in range(n)]

print("\nInitial Specific Hypothesis (S):")

print(S)

print("\nInitial General Hypothesis (G):")

for row in G:

    print(row)

for i, h in enumerate(concepts):

    if target[i].lower() == "yes":

        for x in range(n):

            if h[x] != S[x]:

                S[x] = '?'

                G[x][x] = '?'

    else:

        for x in range(n):

            if h[x] != S[x]:
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        G[x][x] = S[x]

    else:

        G[x][x] = '?'

    print("\nAfter Example", i + 1)

    print("Specific Hypothesis:", S)

    print("General Hypothesis:")

    for row in G:

        print(row)

G = [g for g in G if g != ['?'] * n]

print("\nn=====")

print("Final Specific Hypothesis:")

print(S)

print("\nFinal General Hypothesis:")

for row in G:

    print(row)

```

OUTPUT

```

Training Data:
   Sky AirTemp Humidity   Wind Water Forecast EnjoySport
0  Sunny   Warm   Normal  Strong  Warm   Same      Yes
1  Sunny   Warm   High    Strong  Warm   Same      Yes
2  Rainy   Cold   High    Strong  Warm   Change     No
3  Sunny   Warm   High    Strong  Cool   Change     Yes

Initial Specific Hypothesis (S):
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

Initial General Hypothesis (G):
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

After Example 1
Specific Hypothesis: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
General Hypothesis:
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

```

```
After Example 2
Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
General Hypothesis:
['?' '?' '?' '?' '?' '?']
['?' '?' '?' '?' '?' '?']
['?' '?' '?' '?' '?' '?']
['?' '?' '?' '?' '?' '?']
['?' '?' '?' '?' '?' '?']
['?' '?' '?' '?' '?' '?']

After Example 3
Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']

After Example 4
Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' '?' '?']
General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

=====
Final Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```